



FX PB Collateral

Automating FX Prime Brokerage Monitoring

Replacing tedious manual portal extraction with an automated end-to-end data pipeline and Power BI analytics dashboard



The Business Problem & Enterprise Impact

Understanding the manual bottlenecks in daily FX Prime Brokerage collateral and limits reporting.



Manual Process Challenge

1

Portal Authentication

Operations team had to **manually log into the Prime Brokerage (PB) portal every morning**, navigate through reporting sub-menus, and await slow report loads.

2

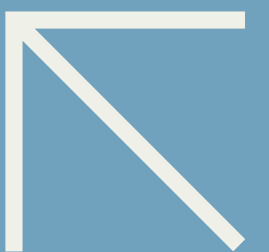
Manual Screenshotting

Capture the relevant **collateral, limits and portfolio information** from the portal reports.

3

Teams Pasting & Delays

Consolidate the extracted information and **manually share updates with stakeholders.**



Implemented Solution

Automated Ingestion Pipeline



Automated Email Delivery

PB portal configured to auto-distribute daily CSV files



Power BI Dataflows

Connects to SharePoint, cleans, aggregates, and structures the raw CSV data for downstream reporting



Power Automate Workflow

Cloud workflow automatically saves CSV files to a secured SharePoint folder



Power BI Reports

Automatically refreshed interactive reports distributed to the primary stakeholders

Operational Speed & Impact

<10s

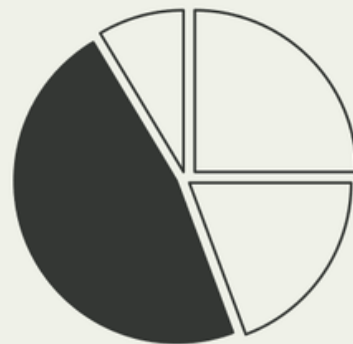
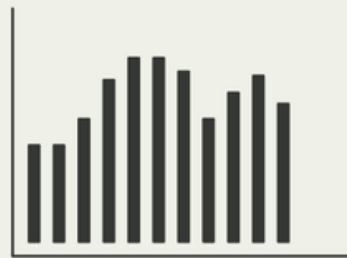
Time to Decision
Insight

Instant Collateral Requirements Visibility

By removing manual portal logins, screenshotting, and Teams messaging, the solution compresses the daily reporting cycle from over **15 minutes** down to **under 10 seconds**.

Stakeholders receive **an updated, reliable collateral report** every morning

Value Added Through Analytics



Power BI Reporting gives additional value

Trend Analysis – Daily collateral movements visualized over time

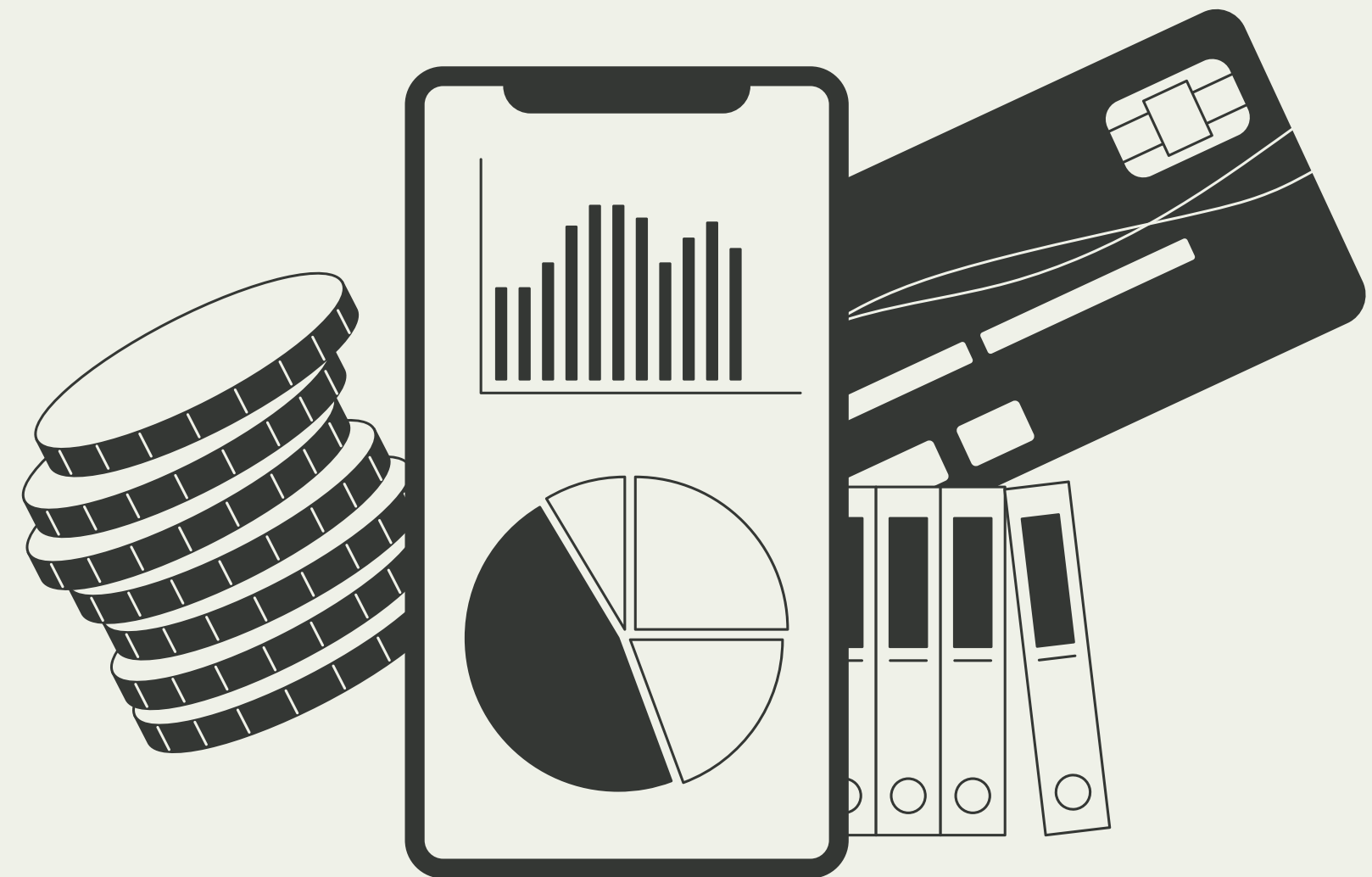
Limit Utilization Monitoring – Daily view of PB limits

Counterparty Exposure Insights – Aggregated exposure by counterparty

Operational Transparency – Stakeholders can self-serve instead of requesting ad-hoc updates.

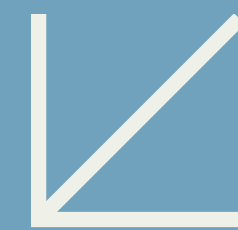
Public Portfolio & GitHub Replication

Enterprise project securely recreated into a public portfolio demonstration





Enterprise vs Public Portfolio



Enterprise Production		GitHub Replication
Prime Brokerage portal CSV feeds	Data	Custom Python script generating synthetic FX data
Microsoft SharePoint Online	Storage	PostgreSQL on Neon Database
Power BI Dataflows to BI reports	Connection	Direct Power BI connection to Neon DB.
Power BI Report	Reporting	BI Tools – Power BI or Google Data Studio

Data Architecture



Python & Neon PostgreSQL Pipeline



Data Generation

Developed a **Python** script to synthesize mock data mimicking original reports



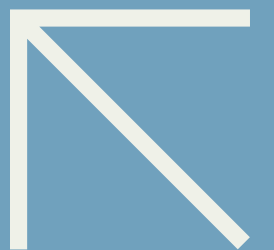
Database Integration

Structured a relational schema and automated data loads directly into a **Neon PostgreSQL** serverless cloud database



Power BI Reports

Connected **Power BI** directly to Neon PostgreSQL database to model data, write DAX measures, and publish the dashboard



AI-Assisted Development

1

Rapid Scripting & Synthetic Logic

Used Generative AI to accelerate the development of complex **Python data-generation scripts**.

2

Database Schema Optimization

Leveraged AI prompts to help design **PostgreSQL tables**—schema layout, primary keys, and data types for Neon PostgreSQL database.

3

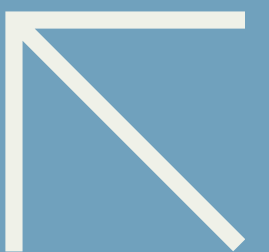
Documentation

Generated docstrings and a comprehensive **GitHub README** explaining setup instructions for reviewers.

4

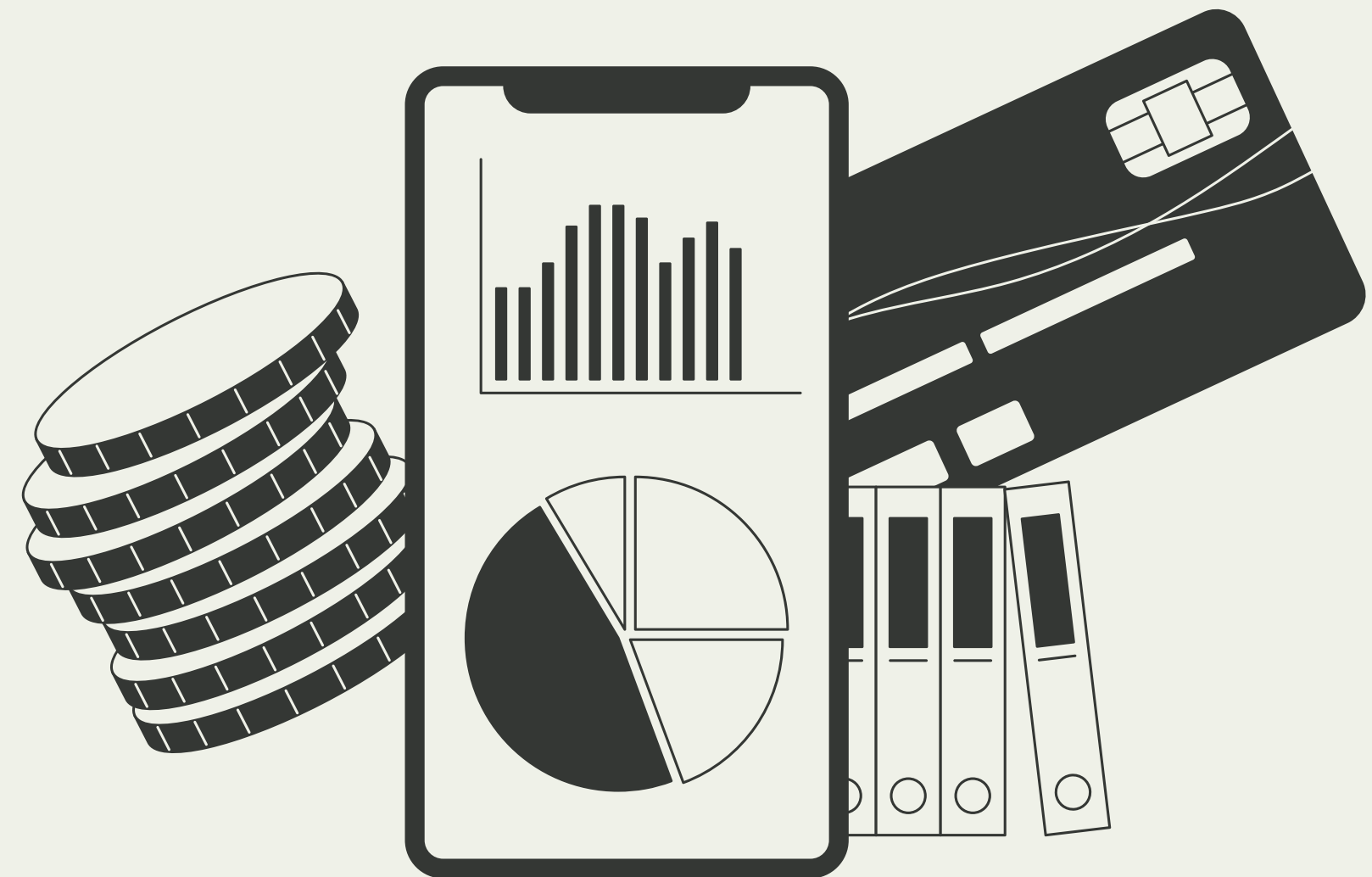
Implementation

AI-accelerated implementation while **designing, testing, and validating** the logic.



Reporting Examples

Interactive Power BI Reports



Collateral Dashboard

Collateral Dashboard

Selected Date
10/Aug/2026

Initial Margin

-5.09M

Change: ▲ 588K

Variation Margin

-46.24K

Change: ▼ 41K

Collateral Required

-5.13M

Change: ▲ 547K

Collateral Posted

6.19M

Change: ▲ 1M

Excess / Deficit

1.06M

Change: ▲ 2M

Total Portfolio (In USD)

102.93M

Change: ▼ 11M

Open Positions

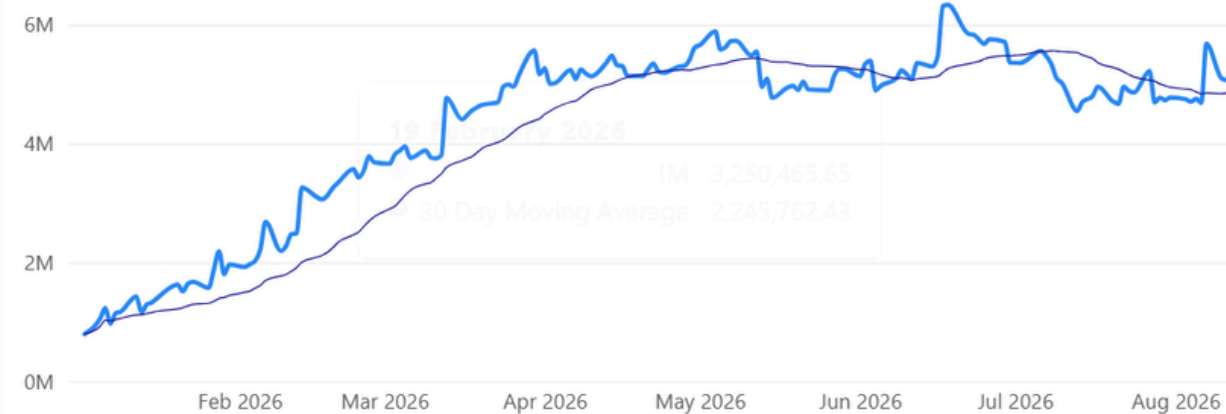
45

Change: ▼ 1

- ☐ Latest Date
- ☒ 10-Aug-2026
- ☐ 07-Aug-2026

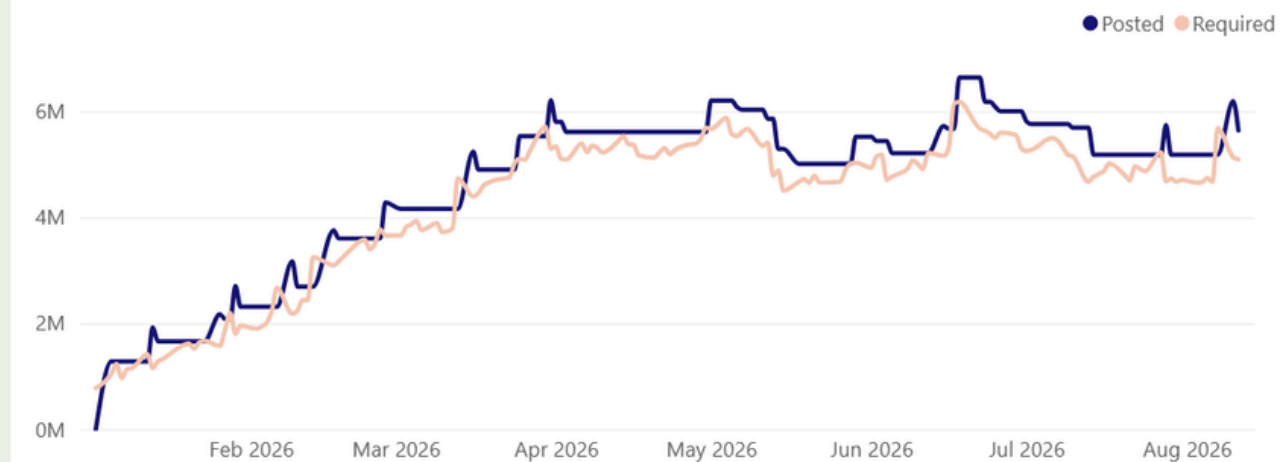
Initial Margin

IM vs 30 Day Moving Average



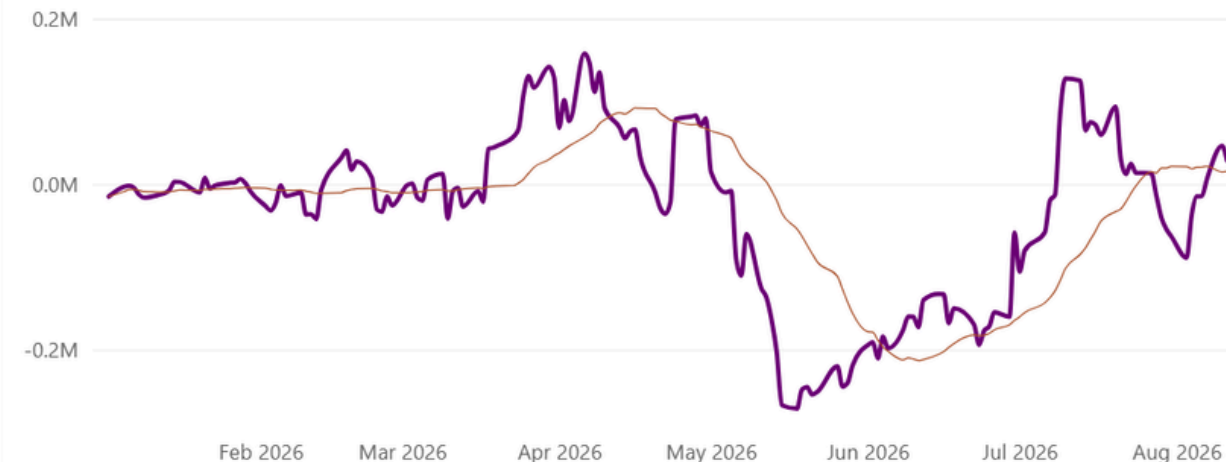
Collateral: Required vs Posted

Posted vs Required

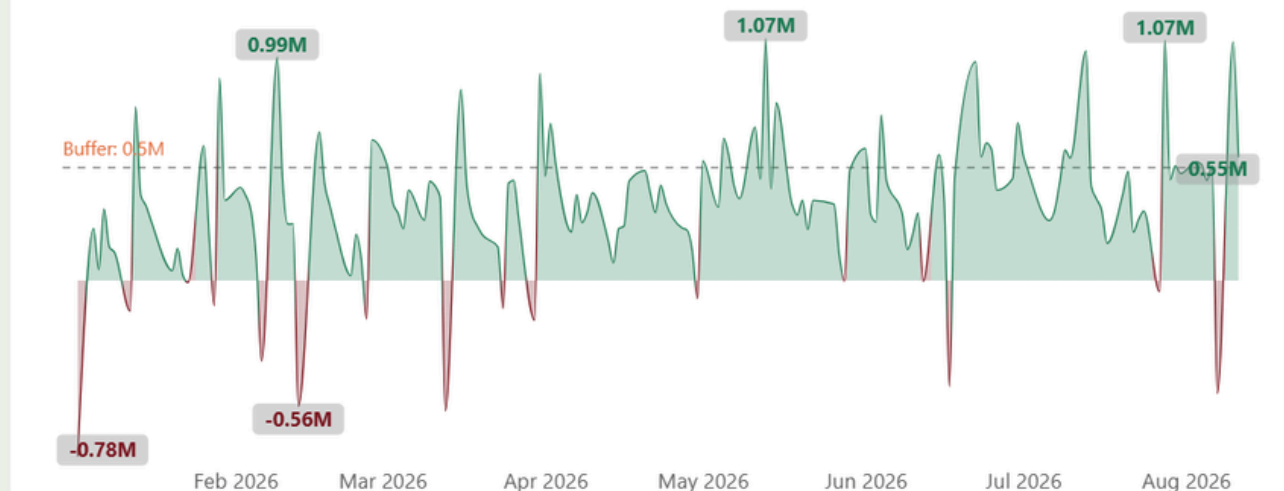


Variation Margin

VM vs 30 Day Moving Average



Excess / Deficit

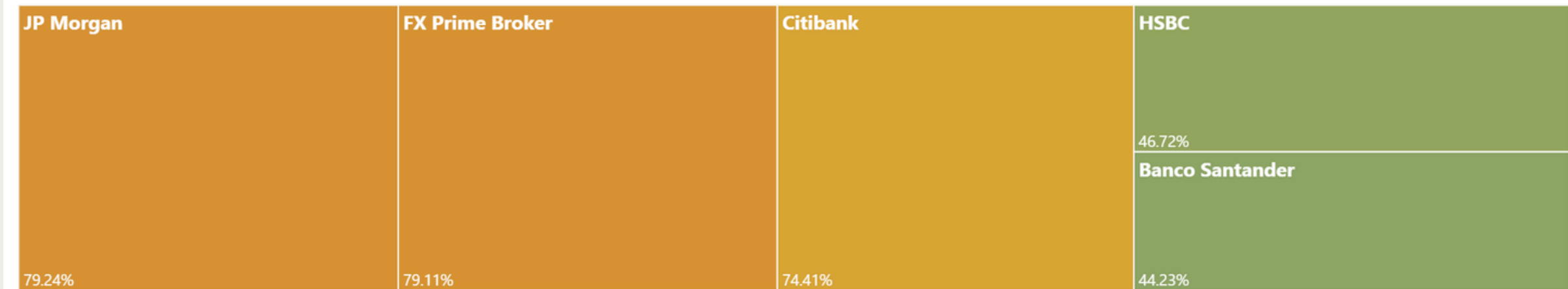


Limits

Limits

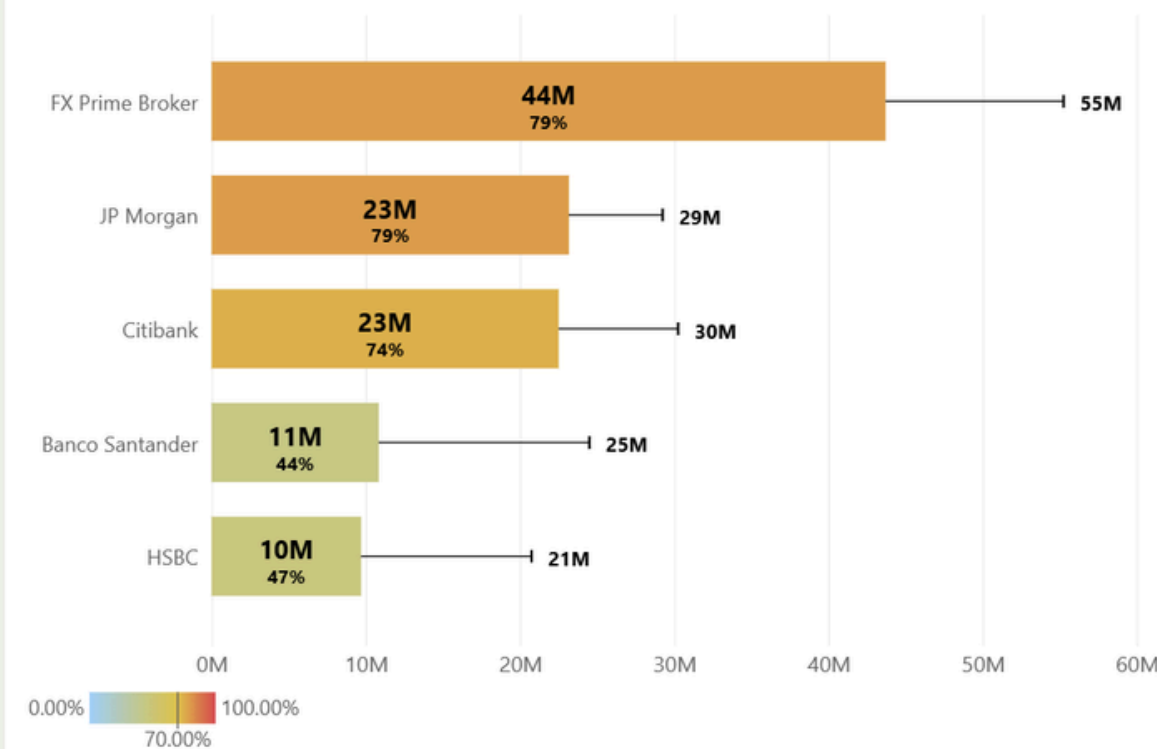
No breaches

2 Banks above 75% of their credit utilization



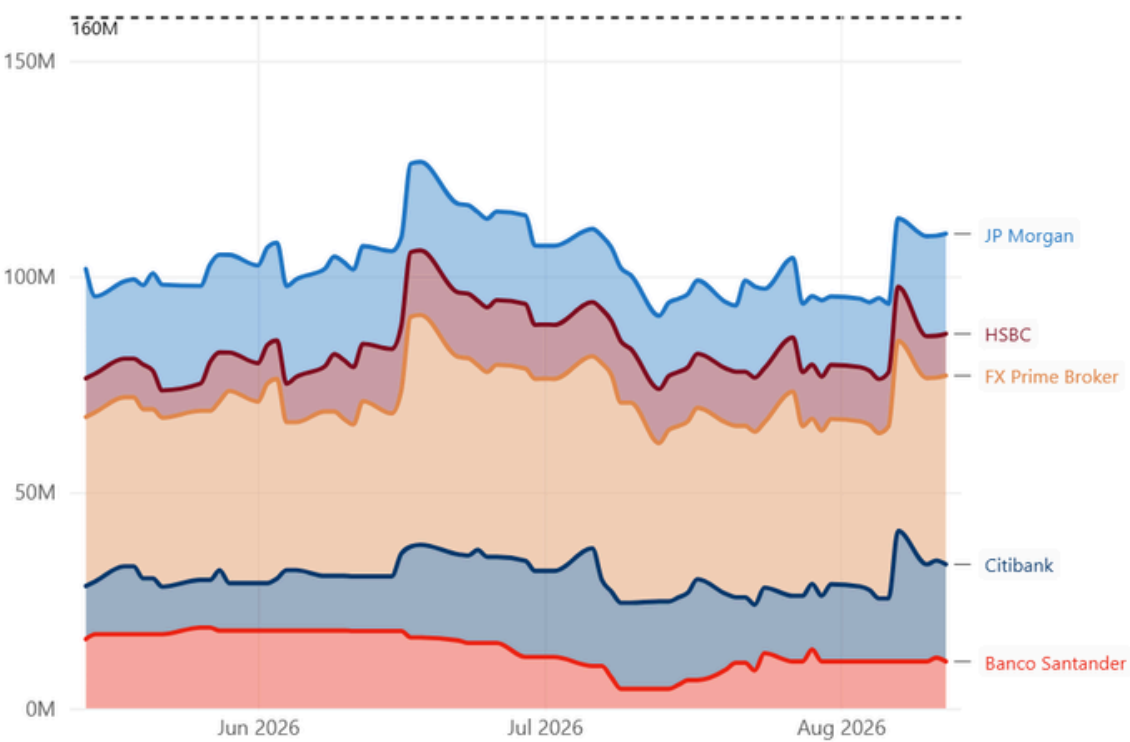
Bank Limits

Limit usage



Portfolio Limits

Over time



Selected Date

12/Aug/2026

☒ Latest Date

☐ 11-Aug-2026

☐ 10-Aug-2026

Select all

Banco Santander

Citibank

FX Prime Broker

HSBC

JP Morgan

Select all

forward

swap

Portfolio

Portfolio

Total Portfolio (In USD)

109.92M

Change: ▲ 458K

Open Positions

45

Change: ▼ 1

Trades

5.08K

Spots

68

Forwards

2.45K

Swaps

2.57K

Selected Date

12/Aug/2026

☒ Latest Date

☐ 11-Aug-2026

☐ 10-Aug-2026

Select all

Banco Santander

Citibank

FX Prime Broker

HSBC

JP Morgan

Select all

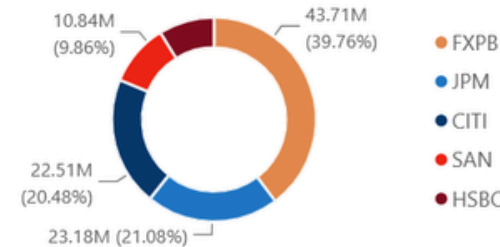
forward

spot

swap

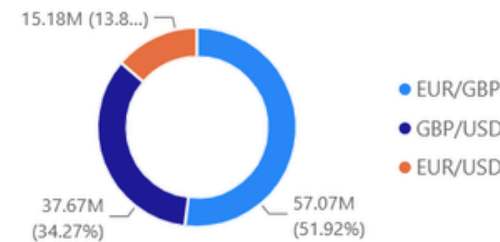
Portfolio Allocation

By Bank



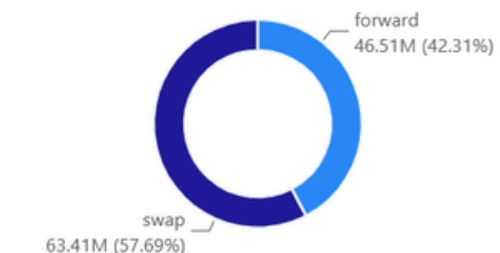
Portfolio Allocation

By Currency Pair

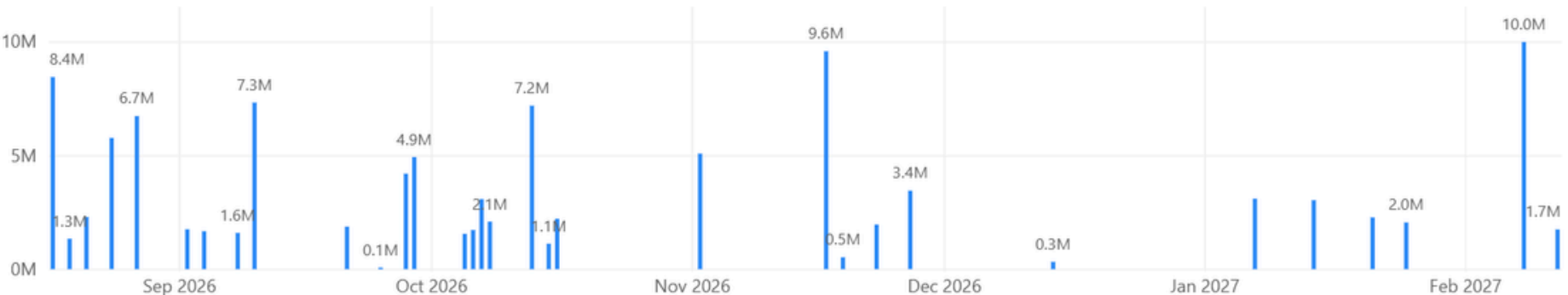


Portfolio Allocation

By type



Maturities



Size in USD



Portfolio

trade_id	type	currency_pair	bank_code	trade_date	value_date	FX Portfolio Size
T00033	swap	EUR/USD	CITI	19 February 2026	19 August 2026	1,332,157.49
T00039	swap	GBP/USD	FXPB	27 February 2026	27 August 2026	2,889,212.77
T00040	forward	EUR/GBP	JPM	02 March 2026	02 September 2026	1,748,368.09
T00053	swap	EUR/GBP	JPM	19 March 2026	21 September 2026	1,867,596.79
T00058	forward	GBP/USD	SAN	26 March 2026	28 September 2026	2,527,846.78
T00059	forward	EUR/GBP	HSBC	27 March 2026	27 August 2026	3,838,135.12
T00066	swap	EUR/USD	FXPB	07 April 2026	07 October 2026	3,074,503.07
T00070	swap	EUR/GBP	CITI	13 April 2026	13 October 2026	2,410,673.40
T00073	forward	GBP/USD	FXPB	16 April 2026	16 October 2026	1,109,000.63
T00082	forward	GBP/USD	FXPB	29 April 2026	29 September 2026	2,025,478.49
T00087	swap	GBP/USD	JPM	06 May 2026	06 October 2026	1,716,280.43
T00094	forward	EUR/USD	SAN	15 May 2026	15 October 2026	1,116,216.96
T00096	forward	GBP/USD	JPM	19 May 2026	19 November 2026	518,842.01
T00098	swap	EUR/GBP	JPM	21 May 2026	21 August 2026	2,288,574.66
Total						109,922,138.08